

WARNING: MISBEHAVIOR AT EXAM TIME WILL LEAD TO SERIOUS CONSEQUENCE.

SCUT Final Exam

Mathematical Analysis I Exam Paper A (2020-2021-1)

- Notice:**
1. Make sure that you have filled the form on the left side of seal line.
 2. Write your answers on **the exam paper**.
 3. This is a **close**-book exam.
 4. The exam with full score of 100 points lasts 120 minutes.

Question No.	I	II	III	IV	V	Sum
Score						

Score

I. Please fill the correct answers in the following blanks. (4' × 6 = 24')

1. If $y = x^{\sqrt{x}} (x > 0)$, then $dy =$ _____.
2. If $\lim_{n \rightarrow \infty} a_n = a$, then $\lim_{n \rightarrow \infty} \frac{[na_n]}{n} =$ _____ where the symbol $[\cdot]$ denotes the greatest integer function.
3. The slant asymptotes of the curve $y = \sqrt{x^2 + 4x}$ are _____.
4. If $f(0) = 0, f'(0) = 1$, then $\lim_{x \rightarrow 0} \frac{f(1 - \cos x)}{\ln(1 - x^2)} =$ _____.
5. The improper integral $\int_1^{+\infty} e^{-x^2} dx$ is (convergent or divergent) _____.
6. $\int_{-1}^1 (x + \sqrt{4 - x^2}) dx =$ _____.

II. Finish the following calculations. (7-11: 6'×5 = 30')

Score

7. $\lim_{x \rightarrow 2} \frac{\sqrt{6-x}-2}{\sqrt[3]{3-x}-1}.$

8. $\lim_{x \rightarrow 0} \frac{\frac{x^2}{2} + 1 - \sqrt{1+x^2}}{(\cos x - e^{-x^2}) \sin x^2}.$

9. $\lim_{n \rightarrow +\infty} \left(\frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{n+n} \right)$

10. If a curve is given by the parametric equation $\begin{cases} x = \ln(t + \sqrt{t^2 + 1}) \\ y = \int_0^t \sin u^2 du \end{cases}, t \in R$, find $\frac{d^2 y}{dx^2}$.

11. Evaluate the indefinite integral $\int x \arcsin x \, dx$.

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III. Applications of calculations. (12-13: 8'× 2 = 16')

Score

12. Find the local maximum and minimum values, the intervals of concavity, and inflection points of the function $f(x) = x^{2/3}(6-x)^{1/3}$.

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13. Find length of one arch of the cycloid $x = a(\theta - \sin \theta)$, $y = a(1 - \cos \theta)$ ($a > 0$ is a constant).

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IV. Prove the following conclusions. (14-16: 7'×3=21')

Score

14. Prove the inequality $\frac{x}{1+x} < \ln(1+x) < x, (x > 0)$.

15. Suppose the function $f(x)$ is continuous on $[0, +\infty)$, $f(0) < 0$, $\lim_{x \rightarrow +\infty} f(x) = A > 0$, show that the equation $f(x) = 0$ has at least one root on $[0, +\infty)$.

16. If $f(x)$ has continuous derivatives up to order 2 on $[a, b]$, and $f'(a) = f'(b) = 0$, show that

there exists $\xi \in [a, b]$ such that

$$|f''(\xi)| \geq \frac{4}{(b-a)^2} |f(b) - f(a)|.$$

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V. Finish the following questions. (17:9'×1=9')

17. Show that the sequence defined by $0 < x_1 < \sqrt{3}$, $x_{n+1} = \frac{3(1+x_n)}{3+x_n}$ ($n = 1, 2, 3, \dots$) is convergent and find its limit.

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